

Chapter 9

Optimal Taxation



The reader interested in studying more extensively optimal taxation can have a look at the book by B. Salanie': *The Economics of Taxation*.

Taxation is a cornerstone of modern economies, providing the resources necessary for public services, infrastructure, and social welfare programs. Yet, designing an income tax system involves navigating complex trade-offs between efficiency and equity. On the one hand, higher taxes on labor income can fund essential public goods and reduce inequality. On the other hand, excessive taxation can distort incentives to work, potentially reducing overall economic

productivity. The theory of optimal income taxation offers a framework for addressing these trade-offs, balancing the goals of redistributive justice and economic efficiency. This chapter explores the principles of optimal labor income taxation through real-world examples and policy challenges.

A clear illustration of these trade-offs can be seen in the progressive income tax systems used in many developed countries. For example, the United States employs a tiered tax system, where individuals earning higher incomes face higher marginal tax rates. This approach aims to reduce income inequality by redistributing wealth from higher earners to fund public programs such as healthcare, education, and social security. However, high marginal tax rates can create disincentives for additional work, particularly for top earners. The case of Sweden, with its historically high tax rates on labor income, demonstrates the balance needed. Despite having some of the highest marginal tax rates globally, Sweden's economy has maintained high labor force participation rates due to complementary policies such as generous childcare support and job training programs, which mitigate work disincentives.

Another compelling example comes from the Earned Income Tax Credit (EITC) in the United States, a program designed to incentivize work among low- and moderate-income individuals. The EITC provides a refundable tax credit that increases with earned income up to a certain threshold, effectively boosting take-home pay for low-wage workers. Studies have shown that the EITC encourages labor force participation, particularly among single mothers. However, the program also introduces potential inefficiencies at higher income levels, where the credit phases out, creating marginal tax rate spikes that may discourage additional work effort.

In contrast, some countries have experimented with flat tax systems, where all labor income is taxed at a single rate. For example, Estonia implemented a flat tax in the 1990s, aiming to simplify the tax system and encourage labor supply. While flat taxes reduce administrative complexity and avoid sharp disincentives at certain income levels, they also limit the scope for redistribution, raising concerns about equity. The trade-off between simplicity and fairness is central to debates over the optimal structure of labor income taxation.

The gig economy provides a modern context for examining labor income taxation. Workers in this sector, such as rideshare drivers or freelance writers, often face irregular income streams and lack traditional employee benefits. Tax systems designed for salaried workers may struggle to account for the unique challenges faced by gig workers. For instance, compliance costs and uncertainty about tax liabilities can deter gig workers from fully reporting their income. Simplified tax schemes, such as the United Kingdom's 'Making Tax Digital' initiative, aim to address these challenges by streamlining reporting requirements, ensuring that labor income taxation remains equitable and efficient in an evolving labor market.

Public debates over the taxation of high earners also highlight the complexities of labor income taxation. Proposals to increase top marginal tax rate - such as Alexandria Ocasio-Cortez's suggestion of a 70% rate on income above \$10 million - spark discussions about the balance between redistribution and economic growth. While such policies can significantly reduce income inequality, critics warn of potential adverse effects on labor supply and tax avoidance behaviors among top earners. Historical evidence from the United States' high tax rates in the mid-20th century suggests that complementary policies, such as deductions for investments or charitable giving, can mitigate these disincentives.

What unites these examples is the universal challenge of designing labor income tax systems that achieve equity and efficiency. Optimal labor income taxation theory provides a framework for understanding these trade-offs, helping policymakers craft tax policies that balance redis-

tributive goals with incentives for productivity and growth.

In this chapter, we will delve into the theoretical foundations of optimal labor income taxation, exploring the trade-offs between equity and efficiency, the role of behavioral responses, and the practical constraints faced by policymakers. By examining real-world applications, readers will gain insights into how labor income tax systems can be designed to achieve socially desirable outcomes in diverse economic contexts.

Roughly speaking, taxation has two main aims:

- Pareto improvement: taxes to improve welfare in the Pareto sense, for example, in order to internalise externalities.
- Redistribution: CE defines an equilibrium but the Pareto planner does not like it, so he uses taxes to redistribute.

We will study the second problem. We saw from the Second Welfare Theorem that, if lump-sum taxes were available, there would be no trade-off between efficiency and redistribution. Unfortunately, lump sum taxes are not always available, for example when there are informational asymmetries. So, the next question becomes of non-trivial solution:

What structure of labor earnings taxation can maximize the social benefits of redistribution and public insurance while minimizing the social harm associated with distorting the allocation of labor input?

There are two approaches to this question in the literature. The Mirrlees approach is to look for tax systems that maximize social welfare subject to the constraint that the planner cannot observe the relative contributions of individual productivity versus individual labor input in producing observed individual earnings. This approach is attractive because it places no constraints on the shape of the tax schedule, and because the implied allocations are constrained efficient. In other terms, the Mirrlees model is a tool to formally study efficiency-equity trade off in shaping an optimal redistributive system.

The alternative Ramsey approach is to restrict the planner to choose a tax schedule within a parametric class, for example, to restrict taxes to be a linear function of earnings. While there are no theoretical foundations for imposing ad hoc restrictions on the design of the tax schedule, the practical advantage of doing so is that one can then consider tax design in richer models. We will not study in detail this second approach however, in certain cases we can see the outcome of a restricted Mirrlees model as Ramsey.

9.1 The Mirrlees Model

We assume that all agents have identical quasi-linear preferences in consumption $c \geq 0$ and labour $\ell \geq 0$. The individual welfare is described by the function $u(c, \ell) = c - v(\ell)$, where $v(\cdot)$ is strictly increasing and convex, $v'(0) = v(0) = 0$. An example of such a function would be the power function $v(\ell) = \frac{\ell^\gamma}{\gamma}$ for $\gamma > 1$. Agents differ in productivity $\theta \in \Theta := \{\theta_1, \theta_2, \dots, \theta_N\}$ and their income depends on both labour and productivity $y = \theta\ell$. To give an order to Θ , we assume: $0 < \theta_1 < \theta_2 < \dots < \theta_N$.

The productivities have a certain distribution in the population $\pi_n > 0$, $n = 1, 2, \dots, N$, and government is unable to observe both productivity and labour supply, it only observes labour income and knows the distribution of productivities. The government can control income and

consumption $\{y_n, c_n\}_{n=1}^N$ through the tax schedule, in order to maximize social welfare subject to the resource feasibility.

The social welfare function is defined as:

$$\sum_{n=1}^N \psi_n \pi_n [c_n - v(\ell_n)]$$

a weighted sum of the utilities for each type. Each utility is weighted by both π_n the frequency of the type in the population and ψ_n that is the (here exogenous) weight assigned to each type θ_n agent by the planner. In order to have an interesting trade off between incentives and redistribution, we assume that $\psi_1 > \psi_2 \geq \dots \psi_n \geq \dots \geq \psi_N \geq 0$. That is, the planner gives, ceteris paribus, more weight to less productive types. How the coefficients ψ_n decrease in n , especially for low n , captures how much redistribution is desirable for the planner. Therefore, it captures how much the planner is willing to give up on efficiency for equality. The assumption on the Pareto weights allows for the most extreme desire for redistribution, the Rawlsian planner: $\psi_1 > 0$ and $\psi_n = 0$ for $n > 1$.

The maximization must be subject to the resource feasibility¹

$$\sum_{n=1}^N \pi_n [y_n - c_n] \geq G,$$

that is, gross income must cover both private C and public consumption G .

For each agent type θ , it will be convenient to set $\ell = \frac{y}{\theta}$ and replace the right hand side into the agent's utility function. We get the new preferences as a function of y : $c - v(\frac{y}{\theta})$. Since $v(\frac{y}{\theta})$ decreases in θ for a fixed y , high productivity is equivalent to lower cost of producing a certain amount of income.

The indifference curves satisfy the Spence-Mirrlees single crossing property. Indeed, given the convexity of v , it is easy to show that for $y' > y$, the difference $v(\frac{y'}{\theta}) - v(\frac{y}{\theta})$ decreases in θ [check it]. This means that for the efficient type increasing y is less costly. As it is displayed in Figure 9.1, in the (y, c) plane, the higher is θ the flatter is the indifference curve, with θ_N having the flattest one.

9.2 First Best Allocation

Let's first consider the case where the productivity θ is observable. We call this case as First Best. The First Best problem solves

$$\begin{aligned} \max_{\{y_n, c_n\}} \quad & \sum_{n=1}^N \psi_n \pi_n \left[c_n - v\left(\frac{y_n}{\theta_n}\right) \right] \\ \text{s.t.} \quad & c_n - v\left(\frac{y_n}{\theta_n}\right) \geq 0, & (IR_n) \quad \forall n \in \{1, 2, \dots, N\} \\ & \sum_{n=1}^N \pi_n [y_n - c_n] \geq G, & (\text{Budget Constraint}) \\ & y_n \geq 0 & (\text{Non Negativity}) \quad \forall n \in \{1, 2, \dots, N\} \end{aligned}$$

¹If the welfare function is the Rawlsian planner, standard dual arguments imply that the solution can be obtained by maximizing revenues subject to giving type $n = 1$ agent at least a pre-specified utility level.

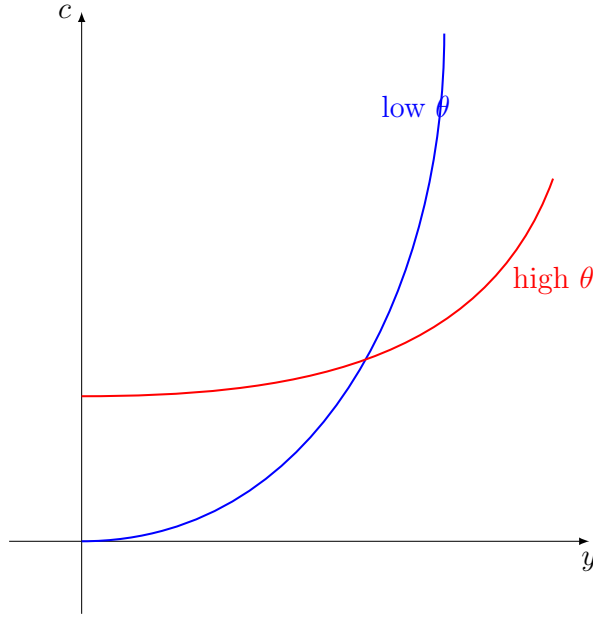


Figure 9.1: Example of indifference curves in the Mirrlees model. The higher the type θ of the agent the flatter the indifference curve: since for higher types it is less costly to increase income the higher the type of the agent the lower the increase in consumption required to maintain the same level of utility (single crossing of indifference curves).

Notice that non-negativity in c_n would be redundant as we already have the *IR* constraints.

Denote by λ the multiplier for BC and ζ_n the one for IR_n . Taking FOCs for interior solutions (assuming also the non-negativity constraint in y_n is slack) we obtain that for each possible n :

$$\begin{aligned} c_n : \psi_n + \frac{\zeta_n}{\pi_n} &= \lambda \\ y_n : \left(\psi_n + \frac{\zeta_n}{\pi_n} \right) v' \left(\frac{y_n^{fb}}{\theta_n} \right) &\geq \lambda \theta_n \quad \text{with equality if } y_n > 0. \end{aligned}$$

Since $\psi_1 > 0$ and $\xi_1 \geq 0$ it must be that $\lambda > 0$, so *the BC must be satisfied with equality*. This result could be also seen by a generalization of our ‘usual’ argument: Suppose we have a feasible program where the constraint is not binding. Then for some positive $\varepsilon > 0$ appropriately small s.t. the constraint remains satisfied we can increase all consumption uniformly: $\tilde{c}_n = c_n + \varepsilon \quad \forall n$. Trivially, IR also remains satisfied and social welfare increases. Hence, the initial program could not have been optimal. It follows that at optimum BC is binding.

By substituting the value of λ obtained from the FOC on c_n into the other condition we see that:

$$v' \left(\frac{y_n^{fb}}{\theta_n} \right) = \theta_n.$$

Note that, given our assumptions on v , we have $y_n > 0$ since $\theta_n > 0$.

Two things are immediately clear. First, all type produce up to the point where their marginal productivity of labor θ_n equals their marginal cost $v'(\ell_n^{fb})$. Hence, everybody is producing efficiently. Moreover, from the properties of v , the output depends on θ and more productivity implies that more quantity produced. The agent with the highest θ_n is the one

who produces the most and but still gets utility $U_N := c_N - v\left(\frac{y_N^{fb}}{\theta_N}\right) = 0$. In fact, as we shall prove below all IR constraints but that of θ_1 are binding at optimum.

Proposition 13. *If $G = 0$, at the optimum, all IR constraints are binding except for θ_1 .*

Proof. First of all, we need to prove that assuming the first constraint as binding leads to absurd consequences. Suppose $\zeta_1 > 0$. Then, from FOCs on c_n and by monotonicity of ψ_n it must be $\zeta_n > 0 \ \forall n$. But then $U_n = 0 \ \forall n$ and thus the value of the objective function must be 0. This is absurd because we can find an allocation where the value of the objective function is strictly positive. This is so since $G = 0$ and $v'(0) = 0$ while $\theta_i > 0$ for all i . [This is left as an exercise]

Hence it must be that $\zeta_1 = 0$. However, by the monotonicity of ψ_n in the types and FOCs on c_n we still get that it must be $\zeta_n > 0 \ \forall n > 1$. $\square$

A graphical example of the first best allocation - in the (y, c) plane - is displayed in Figure 9.2.

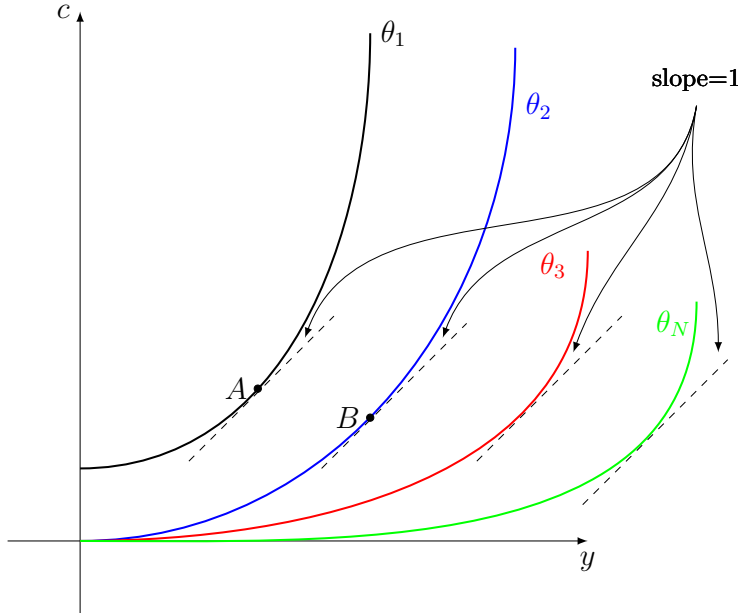


Figure 9.2: Example of first best allocation. At points A and B, the associated indifference curves have unitary slope as required by the efficiency condition for production in the (y, c) plane: $1 = \frac{1}{\theta_n} v'\left(\frac{y_n}{\theta_n}\right)$.

9.3 Second Best

If we assume incomplete information, the allocation must also satisfy the additional family of incentive compatibility constraints. In particular, fixing any possible n , incentive compatibility requires that agent n does not have the desire to pretend to be any other type i . Therefore, we need to consider for each n :

$$IC(n, i) : c_n - v(y_n/\theta_n) \geq c_i - v(y_i/\theta_i),$$

which need to be satisfied for each i .

In order to ease notation, let $LDIC_n := IC(n, n-1)$ be the *local downward* incentive compatibility constraint of the type n , and $LUIC_n := IC(n, n+1)$ be the *local upward* one.

9.3.1 First best and IC

It is easy to see that the First Best allocation is not incentive compatible as long as $U_1 > 0$ or $y_1^{fb} > 0$ (or both), since in this case each higher type has incentive to deviate and to pretend to be type θ_1 . Indeed, consider a generic type $n > 1$ with a binding IR. In the First best allocation, we have:

$$c_1 - v\left(\frac{y_1}{\theta_n}\right) \geq c_1 - v\left(\frac{y_1}{\theta_1}\right) = U_1 \geq 0 = c_n - v\left(\frac{y_n}{\theta_n}\right),$$

where we used $\theta_n > \theta_1$, and the first inequality is strict for $y_1 > 0$.

Exercise Show that when all ψ_n are equal, including ψ_1 (so violating our assumptions), there a first best solution that is incentive compatible, hence optimal for the second best problem as well. Explain intuitively.

9.3.2 Relaxed Problem: Local Monotone Approach

In this section we present a series of results that allows to simplify problems with incentive constraints under single crossing. These results can typically be applied in different contexts.

Proposition 14. *If all LDIC and all LUIC are satisfied then y must be monotone non decreasing in n :*

$$y_n \geq y_{n-1} \quad \forall n > 1.$$

Proof. Fix any n s.t. $1 < n \leq N$. Consider $LUIC_{n-1}$ and $LDIC_n$. If they are satisfied, then we can sum them over and obtain:

$$c_n - v(y_n/\theta_n) + c_{n-1} - v(y_{n-1}/\theta_{n-1}) \geq c_{n-1} - v(y_{n-1}/\theta_n) + c_n - v(y_n/\theta_{n-1})$$

After eliminating consumption from both sides, and a simple rearrangement, we obtain a necessary condition for IC:

$$v(y_n/\theta_{n-1}) - v(y_n/\theta_n) \geq v(y_{n-1}/\theta_{n-1}) - v(y_{n-1}/\theta_n).$$

Suppose now per contra that $y_n < y_{n-1}$. By the convexity of $v(\cdot)$, and since $\theta_n > \theta_{n-1}$, we would obtain that:

$$v(y_n/\theta_{n-1}) - v(y_n/\theta_n) < v(y_{n-1}/\theta_{n-1}) - v(y_{n-1}/\theta_n),$$

which would contradict the statement above and hence cannot be IC. $\square$

Lemma 9. *If all LDIC and all LUIC are satisfied then all upward IC constraints are satisfied.*

Proof. Part 1: From the fact that LUIC and LDIC are satisfied we get monotonicity in quantities, as shown above. Now fix any n such that $1 \leq n < N-1$ (otherwise there is nothing to show). Consider $LUIC_n$ and $LUIC_{n+1}$, we want to show that $IC(n, n+2)$ is satisfied. Summing over and rearranging the two LUIC one gets:

$$c_n - v(y_n/\theta_n) \geq c_{n+2} - v(y_{n+2}/\theta_{n+1}) + v(y_{n+1}/\theta_{n+1}) - v(y_{n+1}/\theta_n). \quad (9.1)$$

Moreover, by monotonicity of y and the convexity of v we get that:

$$v(y_{n+2}/\theta_n) - v(y_{n+2}/\theta_{n+1}) \geq v(y_{n+1}/\theta_n) - v(y_{n+1}/\theta_{n+1}),$$

which implies that:

$$v(y_{n+2}/\theta_n) \geq v(y_{n+1}/\theta_n) - v(y_{n+1}/\theta_{n+1}) + v(y_{n+2}/\theta_{n+1}). \quad (9.2)$$

It we now combine (9.1) with (9.2) we get:

$$c_n - v(y_n/\theta_n) \geq c_{n+2} - v(y_{n+2}/\theta_n),$$

which is exactly what we wanted to show.

Part 2: Now we proceed with the inductive argument to show that $IC(n, k)$ holds assuming $IC(n, k-1)$ and $k = n+2, n+3, \dots, N$. Consider $LUIC_{k-1}$ and $IC(n, k-1)$, sum them over and rearrange to obtain:

$$c_n - v(y_n/\theta_n) \geq c_k - v(y_k/\theta_{k-1}) + v(y_{k-1}/\theta_{k-1}) - v(y_{k-1}/\theta_n)$$

As before use monotonicity of quantities and convexity of cost function to obtain that:

$$v(y_k/\theta_n) \geq v(y_k/\theta_{k-1}) - v(y_{k-1}/\theta_{k-1}) + v(y_{k-1}/\theta_n)$$

Which in pair with the inequality above gives:

$$c_n - v(y_n/\theta_n) \geq c_k - v(y_k/\theta_n).$$

□

Lemma 10. *If all $LUIC$ and $LDIC$ are satisfied then also all downward IC constraints are satisfied.*

Proof. Very similarly to before. Fix any $2 < n \leq N$, otherwise there is nothing to prove. Part 1: We want to show that $IC(n, n-2)$ is satisfied. consider $LDIC_n$ and $LDIC_{n-1}$, summing them over and rearranging one gets:

$$c_n - v(y_n/\theta_n) \geq c_{n-2} - v(y_{n-2}/\theta_{n-1}) + v(y_{n-1}/\theta_{n-1}) - v(y_{n-1}/\theta_n)$$

From convexity and monotonicity as usual we get:

$$v(y_{n-2}/\theta_{n-1}) - v(y_{n-2}/\theta_n) \leq v(y_{n-1}/\theta_{n-1}) - v(y_{n-1}/\theta_n).$$

Finally:

$$v(y_{n-2}/\theta_n) \geq v(y_{n-2}/\theta_{n-1}) - v(y_{n-1}/\theta_{n-1}) + v(y_{n-1}/\theta_n).$$

Therefore, together with first inequality we get to:

$$c_n - v(y_n/\theta_n) \geq c_{n-2} - v(y_{n-2}/\theta_n)$$

Which is the IC we wanted to show.

Part 2: Similarly to the previous claim we can set up an inductive argument to also show that all the other DIC are satisfied. The argument is very similar to the previous one, just notationally more tedious. Assume $IC(n, n-(k-1))$ is satisfied we want to show also $IC(n, n-k)$ is satisfied (for any

k s.t. $n - k \geq 1$). Consider $IC(n, n - (k - 1))$ and $LDIC_{n-k+1}$. Summing them over and making the usual rearrangements:

$$c_n - v(y_n/\theta_n) \geq c_{n-k} - v(y_{n-k}/\theta_{n-k+1}) + v(y_{n-k+1}/\theta_{n-k+1}) - v(y_{n-k+1}/\theta_n).$$

From convexity of costs and monotonicity of output and productivity:

$$v(y_{n-k}/\theta_{n-k+1}) - v(y_{n-k}/\theta_n) \leq v(y_{n-k+1}/\theta_{n-k+1}) - v(y_{n-k+1}/\theta_n).$$

Thus we get that:

$$v(y_{n-k}/\theta_n) \geq v(y_{n-k}/\theta_{n-k+1}) - v(y_{n-k+1}/\theta_{n-k+1}) + v(y_{n-k+1}/\theta_n)$$

Hence, combined with the result above:

$$c_n - v(y_n/\theta_n) \geq c_{n-k} - v(y_{n-k}/\theta_n)$$

Which completes the proof. $\square$

From the two lemmas above we straightforwardly obtain the following corollary.

Corollary 2. *Necessary and sufficient conditions for the (global) incentive compatibility is that local downward and local upward constraints are satisfied.*

This local-global result suggest that the agent's problem is concave when we assume single crossing.

One approach is hence to solve the problem only considering local constraints. In contexts like ours, where the trade offs are quite transparent, another simplification is possible, summarised by the next proposition.

Proposition 15. *If all LDIC bind and income is monotone then all LUIC are satisfied.*

Proof. Fix any n s.t. $n < N$. From $LDIC_{n+1}$ we get that:

$$c_{n+1} - c_n = v(y_{n+1}/\theta_{n+1}) - v(y_n/\theta_{n+1}).$$

Rearranging $LUIC_n$ we obtain that it is satisfied iff:

$$c_{n+1} - c_n \leq v(y_{n+1}/\theta_n) - v(y_n/\theta_n).$$

Thus, this is what we want to show. Exploiting monotonicity in output and convexity of costs as usual we get:

$$v(y_n/\theta_n) - v(y_n/\theta_{n+1}) \leq v(y_{n+1}/\theta_n) - v(y_{n+1}/\theta_{n+1}).$$

Rearranging and combining with the binding LDIC, we get (as desired):

$$c_n - c_{n-1} = v(y_{n+1}/\theta_{n+1}) - v(y_n/\theta_{n+1}) \leq v(y_{n+1}/\theta_n) - v(y_n/\theta_n).$$

$\square$

A second approach is hence to only consider the local downward constraints and impose monotonicity and then show that at the optimum the LDIC are binding. This is the approach typically used by the literature (with discrete types) and what we will follow next.

9.3.3 Constrained Maximization Problem

Given what we have shown in the previous section, the relaxed Second Best problem solves

$$\begin{aligned}
& \max_{\{y_n, c_n\}_{n=1}^N} \sum_{n=1}^N \pi_n \left[\psi_n \left(c_n - v \left(\frac{y_n}{\theta_n} \right) \right) + \lambda (y_n - c_n) \right] - \lambda G \quad (9.3) \\
& s.t. \quad c_n - v \left(\frac{y_n}{\theta_n} \right) \geq c_{n-1} - v \left(\frac{y_{n-1}}{\theta_n} \right), \quad (LDIC_n) \quad \forall n = 2, \dots, N; \\
& \quad y_n \geq y_{n-1} \quad (M_n) \quad \forall n = 2, \dots, N; \\
& \quad c_n - v \left(\frac{y_n}{\theta_n} \right) \geq 0 \quad (IR_n) \quad \forall n = 1, 2, \dots, N.
\end{aligned}$$

In the above formulation, we have already inserted the budget constraint (with multiplier λ) into the objective function. This is again common practice in the optimal taxation literature where λ represents the social price of resources. Of course, one should not forget that λ is endogenous and that it is associated to the complementary slackness condition.

From $LDIC_n$ and M_n it is clear that c_n will be increasing in n . We will start to solve the problem by *ignoring the monotonicity constraints* M_n . We will check them ex post.

There are three main approaches to solve problem (9.3): the Kuhn-Tucker method, the free maximization after having studied the constraints in detail, and finally the use of Euler Perturbations (which is in fact the principle behinds both previous approaches).

Kuhn-Tucker. Let us first consider the Kuhn-Tucker approach. Construct the Lagrangian using μ_i for $LDIC_i$ and ζ_i for IR_i . Taking FOCs.

$$\begin{aligned}
c_n : \quad & \psi_n + \frac{\zeta_n + \mu_n}{\pi_n} = \lambda + \frac{\mu_{n+1}}{\pi_n}. \\
y_n : \quad & v' \left(\frac{y_n^*}{\theta_n} \right) \left[\psi_n + \frac{\zeta_n + \mu_n}{\pi_n} \right] = \lambda \theta_n + \frac{\mu_{n+1}}{\pi_n} \frac{\theta_n}{\theta_{n+1}} v' \left(\frac{y_n^*}{\theta_{n+1}} \right).
\end{aligned}$$

Where $\mu_1 = \mu_{N+1} = 0$ because the corresponding constraints do not exist.

First of all, it is easy to show that at the second best allocation, $\lambda > 0$ so the budget constraint must be binding. The proof is entirely similar to the First Best case because *a uniform increment in consumption for all types will not affect IC compatibility*. [check it!]

Now, rearranging the FOCs we have

$$\lambda \left[v' \left(\frac{y_n^*}{\theta_n} \right) - \theta_n \right] = \frac{\mu_{n+1}}{\pi_n} \left[\frac{\theta_n}{\theta_{n+1}} v' \left(\frac{y_n^*}{\theta_{n+1}} \right) - v' \left(\frac{y_n^*}{\theta_n} \right) \right].$$

We hence obtain the key result of this model. If $\mu_{n+1} > 0$, since $\frac{\theta_n}{\theta_{n+1}} v' \left(\frac{y_n^*}{\theta_{n+1}} \right) < v' \left(\frac{y_n^*}{\theta_n} \right)$, we have $v' \left(\frac{y_n^*}{\theta_n} \right) < \theta_n$; and, since $\mu_{N+1} = 0$, then $v' \left(\frac{y_N^*}{\theta_N} \right) = \theta_N$. That is, as a generalization to the two type case we have *downwards distortion* for types $n < N$ and *no distortion at the top* (i.e., for agent N) with respect to the first best scenario.

To see the sign and magnitude of the multiplier, notice that if we assume $\zeta_n = 0$ for all n , from the FOCs w.r.t c_n we have

$$\psi_n + \frac{\mu_n}{\pi_n} = \lambda + \frac{\mu_{n+1}}{\pi_n}.$$

or

$$\mu_{n+1} = \mu_n + \pi_n[\psi_n - \lambda].$$

Since $\mu_1 = \mu_{N+1} = 0$ we have both

$$\mu_{n+1} = \sum_{s=1}^n \pi_s[\psi_s - \lambda] \quad \text{and} \quad \mu_n = \sum_{i=n}^N \pi_i[\lambda - \psi_i].$$

Recalling the meaning of the Kuhn-Tucker multiplier, the last equation indicates that the social value for relaxing the IC constraint for agent n depends on the relative (to the social value for resources λ) Pareto weight of all agents above n .

Figure 9.3 reports a graphical example of a second best allocation for three types. Recall the single-crossing property of the indifference curves. The areas where we are able to separate the different types are those between the indifference curves for θ_1 and θ_2 and the one between the indifference curves of θ_2 and θ_3 represented in the figure. Since the *IC* for agent θ_2 is binding, he must be indifferent between his contract B and the one designed for agent θ_1 (denoted by A), similarly for agent θ_3 , who must be indifferent between B and C .

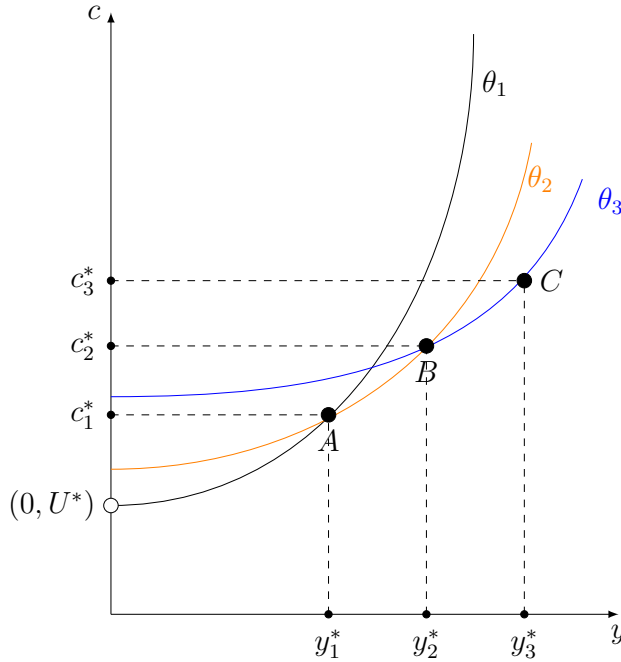


Figure 9.3: Example of a Second Best allocation with three types. Since in the second best, the *IC* for agent θ_2 is binding, he must be indifferent between his contract B and the one designed for agent θ_1 (denoted by A), similarly for agent θ_3 , who must be indifferent between B and C .

9.3.4 Implementation of Optimal Taxation

In this section we ask an important practical question associated to the nature of the application we are considering. Can the second best allocation be obtained using a less theoretical approach than mechanism design? In particular, instead of thinking of a planner that asks to each single agent its unobservable productivity and s/he tells the truth because of IC, can the

planner/government design a tax schedule based on what he sees: gross income y and design such tax schedule in a way to induce each type θ_n to choose the second best quantities y_n^* and obtain the consumption level c_n^* ?

Within a taxation interpretation, consumption c_n^* can be seen as net income, so the tax/transfer scheme must solve:

$$c_n^* = y_n^* - T_n^*.$$

In this problem, we assume that each individual - regardless of his type - faces the same tax schedule $T : Y \rightarrow \mathbb{R}$ with $Y \subset \mathbb{R}_+$, and solves

$$\begin{aligned} \max_{c, y} \quad & c - v\left(\frac{y}{\theta}\right) \\ \text{s.t.} \quad & c \leq y - T(y), \quad y \geq 0. \end{aligned}$$

The budget constraint is obviously binding and so the problem becomes

$$\max_{y \geq 0} \quad y - T(y) - v\left(\frac{y}{\theta}\right).$$

From this maximization problem, taking the FOC, we get the expression for the optimal marginal tax for $y > 0$:

$$T'(y) = 1 - \frac{1}{\theta} v'\left(\frac{y}{\theta}\right).$$

If we implement the second best allocation, for each θ_n we must have $y = y_n^*$ and so marginal taxes turn to be positive for everyone, except for the top guy, for which they are zero. Clearly, higher the marginal taxes, lower the incentive to earn more.

We hence already make the following observation. (i) The marginal tax rate designed in this way is *positive* but it cannot be always increasing in y_n as it must be zero for $n = N$. The optimal tax scheme coming from this problem must have some segments in which it is ‘regressive’. In particular, the model implies that we cannot have negative marginal taxes (such as, for example, the one we observe for low income individuals in the Earned Income Tax Credit program in the US).

As we said, we cannot have increasing *marginal tax* along all income segments. Note however that since the second best schedule typically implies an initial positive transfer, we cannot exclude that *average tax* $\frac{T(y_n)}{y_n} = \frac{y_n - c_n}{y_n}$ increases with income along all income segments.

The Taxation Principle But let’s go back to the original question: Can we implement the second best allocation with such a decentralized instrument such as an income tax schedule T , where each individual of type θ faces the same tax schedule and solves the following problem?

$$\max_{y \geq 0} \quad y - T(y) - v\left(\frac{y}{\theta}\right).$$

The answer is positive as the next proposition shows.

Proposition 16. *For each allocation $\{(c_n^*, y_n^*)\}_{n=1}^N$ that solves the IC constraints, we can construct a tax schedule $T(\cdot)$ such that any individual with skill θ_n optimally chooses the pair (c_n^*, y_n^*) .*

Proof. The incentive compatibility constraint can be rewritten as

$$y_n - T(y_n) - v\left(\frac{y_n}{\theta_n}\right) \geq y - T(y) - v\left(\frac{y}{\theta_n}\right) \quad \forall y \in Y. \quad (IC_y)$$

We can set a tax schedule $T(y_n^*) = y_n^* - c_n^*$ for all y for which there is a θ_n such that $y = y_n^*$. For all other y 's, we set $T(y) = \infty$. This way, we punish in an infinite way the choices of y incompatible with the second best allocation. Incentive compatibility of the original mechanism design problem hence guarantees (IC_y) for all types. $\square$

With the tax schedule, we can target each type, just by targeting his/her y . In the literature this exercise is indicated as *decentralization of the optimal allocation*, in the sense that when facing the common tax schedule $T(\cdot)$, each agent independently decides to choose the level of income designed for him.

If you are unhappy with the infinite punishment note that: (i) as we will see next, this is not needed. (ii) Moreover, when set of types Θ is a continuum, except for pathological situations, there is an interval $Y := [0, y_{max}]$ such that for all $y \in Y$ there is a $\theta \in \Theta$ that is required to produce y in the second best allocation. That is, with a continuum of types, all y in a given interval are admissible and incentive compatibility pins down both total and marginal tax. We will not consider the continuum of types however. There is nothing deep to be learned from this case. We do have a few practical advantages, at the cost of adding un-necessary technicalities.

Implementation with Lower envelope inducing taxes In Figure 9.4, we build on the graphical representation of the second best allocation in Figure 9.3, and show that in fact, the consumption schedule $c(y) := y - T(y)$ that supports the second best allocation can be quite different from the one using infinite punishments such as the proof of Proposition 16 would suggest.

Formally, the implementation displayed in the figure can be described as follows. Let, for $n = 1, 2, \dots, N$, $u_n^* := c_n^* - v\left(\frac{y_n^*}{\theta_n}\right)$ the utility guaranteed to agent θ_n by the targeted allocation.

Recall, in order to implement the targeted allocation, the tax schedule T must be such that for each θ_n :

$$y - T(y) - v\left(\frac{y}{\theta_n}\right) \leq u_n^*.$$

This for all $y \geq 0$ with equality for $y = y_n^*$. This is equivalent to requiring:

$$\begin{aligned} T(y) &\geq T^*(y) & \forall y \\ T(y) &= T^*(y) & \forall y = y_n^*, n = 1, 2, \dots, N \end{aligned}$$

where

$$T^*(y) := \max_n \left\{ y - u_n^* - v\left(\frac{y}{\theta_n}\right) \right\}.$$

It is clear that any tax schedule T that implements the targeted allocation cannot be differentiable at the points $y = y_n^*$. As shown in the figure, at those points each one of such schedules have a convex kink.²

It is however possible to formulate the following conjecture:

²Formally, the left derivative is smaller than the right derivative whenever they exist.

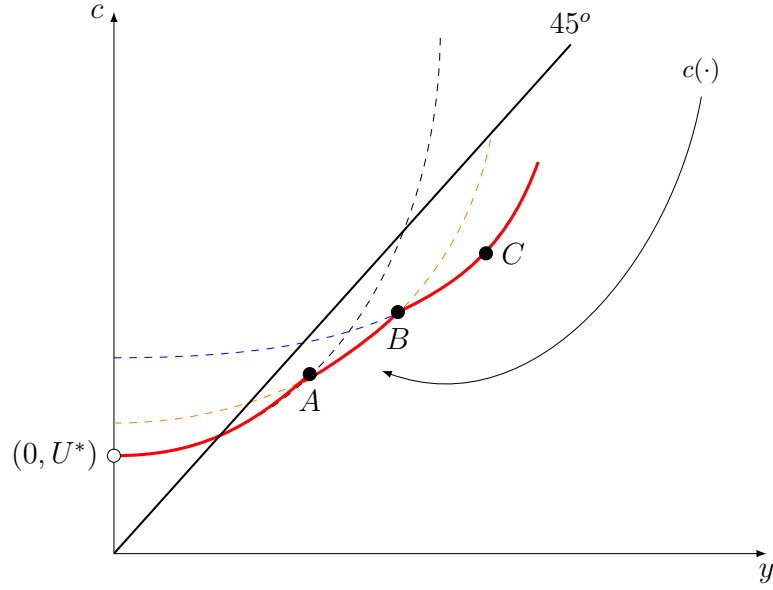


Figure 9.4: Graphical Representation of the Optimal Contract. If consumption follows the thick red envelope of the indifference curves, call it $c(\cdot)$ the agent θ_1 would find it (weakly) optimal to choose A , agent θ_2 finds it optimal to choose B and so on. In other terms, the tax schedule $T(y) := y - c(y)$ ‘implements’ the second best allocation in this case.

Conjecture 1. *Take an incentive compatible allocation $\{c_n^*, y_n^*\}_{n=1}^N$. For each $\varepsilon > 0$, we can construct a differentiable tax schedule $T^\varepsilon(\cdot)$ that implements an incentive compatible allocation $\{c_n^\varepsilon, y_n^\varepsilon\}_{n=1}^N$ such that for all $n = 1, 2, \dots, N$, $|c_n^\varepsilon - c_n^*| < \varepsilon$ and $|y_n^\varepsilon - y_n^*| < \varepsilon$.*

The idea is to perturb the allocation a bit so that the LDIC constraint is not binding, but satisfied with strict inequality (while keeping the LUIC still satisfied). We will hence loose a bit of welfare. But this is conjectured to be arbitrarily small as $\varepsilon \rightarrow 0$ (see section 3.3 in an earlier (2010) version of Ivan Werning’s working paper entitled: ‘Nonlinear capital taxation’).

9.3.5 Substitution and Free Maximization Approach

Let’s go back to the mechanism design approach. In this section, we solve the second best problem (9.3), by following the approach that we adopted in the adverse selection model with two types: Before taking the first order conditions, we guess which constraints are binding and which ones are slack. Then we replace in the objective function the binding constraints. This way we will be able to eliminate the consumption choices and we will be able to take a substantially *free maximisation* over the quantities $\{y_n\}_{n=1}^N$ alone.

Claim 16. *The IR_n are redundant for all $n > 1$.*

Proof. Since the lowest type must obtain at least zero utility, the incentive compatibility constraint implies that the individual rationality constraint is satisfied for any other type. Indeed, consider $LDIC(n, n-1)$

$$U_n = c_n^* - v\left(\frac{y_n^*}{\theta_n}\right) \geq c_{n-1}^* - v\left(\frac{y_{n-1}^*}{\theta_n}\right) \geq c_{n-1}^* - v\left(\frac{y_{n-1}^*}{\theta_{n-1}}\right) \geq U_{n-1},$$

with strict inequality when $y_{n-1}^* > 0$. If we set $n = 2$ and use IR_1 , the previous chain of inequalities imply IR_n for $n > 1$ as claimed. $\square$

Claim 17. *At least one solution to the relaxed problem (9.3) has $LDIC_n$ (for $n > 1$) binding. If ψ_n is strictly decreasing then any solution to (9.3) must have $LDIC_n$ binding.*

Proof. Suppose per contra that there exists a (feasible) solution to the problem where LDIC is not binding for some n . Call such contract as $\{c_n, y_n\}_{n=1}^N$. Consider now a marginal variations in c_n and c_{n-1} alone, that is, without any change in the production quantities $\{y_n\}_n$. Since $LDIC_n$ is slack we can reduce c_n and increase c_{n-1} by a small amount and still have LDIC feasibility. In particular, set:

$$\tilde{c}_n = c_n - \pi_{n-1}\varepsilon,$$

$$\tilde{c}_{n-1} = c_{n-1} + \pi_n\varepsilon,$$

for $\varepsilon > 0$ small enough. By construction, there is no net effect on the BC because the two variations cancel each other out when weighted by the respective π 's. The IR is still satisfied for a marginal variations by the previous lemma where we showed redundancy of IR constraints. $LDIC_{n+1}$ and $LDIC_{n-1}$ are satisfied a fortiori and all other $LDIC$ are not affected. Thus our variation of consumption is feasible. Recall indeed that the relaxed problem does not contemplate upwards IC constraints.

The negative effect to the welfare of the reduction for the θ_n type consumption is:

$$-\psi_n\pi_n\pi_{n-1}\varepsilon,$$

while the positive welfare effect due to the increase for the low type is:

$$\psi_{n-1}\pi_n\pi_{n-1}\varepsilon.$$

Summing up the two terms, we get that the net effect is nonnegative because $\psi_n \leq \psi_{n-1}$ by assumption, and the net effect is positive whenever the inequality between the Pareto weights is strict. $\square$

Let's now make use of the LDIC binding to eliminate consumption. First, for agent θ_1 , by the definition of U_1 , we have:

$$c_1 = U_1 + v\left(\frac{y_1}{\theta_1}\right).$$

Then, using the $LDIC_2$ binding, we have:

$$U_2 = c_2 - v\left(\frac{y_2}{\theta_2}\right) = c_1 - v\left(\frac{y_1}{\theta_2}\right) = U_1 + v\left(\frac{y_1}{\theta_1}\right) - v\left(\frac{y_1}{\theta_2}\right);$$

and so on, using recursively the binding $LDIC$ s, we get, for $n > 1$:

$$U_n = c_n - v\left(\frac{y_n}{\theta_n}\right) = U_1 + \underbrace{\sum_{s=1}^{n-1} \left[v\left(\frac{y_s}{\theta_s}\right) - v\left(\frac{y_s}{\theta_{s+1}}\right) \right]}_{\text{information rents}}$$

Note that type θ_n , gets utility larger than θ_1 due to cumulation of information rents.

The objective function of the relaxed problem becomes

$$\begin{aligned}
& \sum_{n=1}^N \pi_n \left\{ \psi_n U_n + \lambda \left[y_n - \underbrace{\left(U_n + v\left(\frac{y_n}{\theta_n}\right) \right)}_{c_n} \right] \right\} - \lambda G \\
& \sum_{n=1}^N \pi_n \left\{ (\psi_n - \lambda) U_n + \lambda \left[y_n - v\left(\frac{y_n}{\theta_n}\right) \right] \right\} - \lambda G \\
& \sum_{n=1}^N \pi_n \left\{ (\psi_n - \lambda) \underbrace{\left[U_1 + \sum_{s=1}^{n-1} \left(v\left(\frac{y_s}{\theta_s}\right) - v\left(\frac{y_{s+1}}{\theta_{s+1}}\right) \right) \right]}_{U_n} + \lambda \left[y_n - v\left(\frac{y_n}{\theta_n}\right) \right] \right\} - \lambda G.
\end{aligned}$$

The expression can be further rearranged and rewritten as³

$$\sum_{n=1}^N \left\{ (1 - P_n) \mathbf{E}[\psi - \lambda | s > n] \left[v\left(\frac{y_n}{\theta_n}\right) - v\left(\frac{y_{n+1}}{\theta_{n+1}}\right) \right] + \pi_n \lambda \left(y_n - v\left(\frac{y_n}{\theta_n}\right) \right) \right\} + U_1 \mathbf{E}[\psi - \lambda],$$

where: $P_n := \sum_{s=1}^n \pi_s$ and $\mathbf{E}[\psi - \lambda | s > n] := \frac{\sum_{s=n+1}^N \pi_s (\psi_s - \lambda)}{1 - P_n}$. We also assumed $G = 0$ for simplicity.

First of all consider, the FOC on U_1 :

$$\begin{cases} \mathbf{E}(\psi - \lambda) = 0 & \text{if } U_1 > 0 \\ \mathbf{E}(\psi - \lambda) \leq 0 & \text{if } U_1 = 0 \end{cases}$$

Note moreover, that by monotonicity of the Pareto weights we have:

Proposition 17. *We have:*

$$\mathbf{E}[\psi - \lambda | s > n] < \mathbf{E}[\psi - \lambda] \leq 0 \quad \forall n.$$

This result will prove useful below.

³The re-arrangement of the first term is similar to an integration by parts. Here are some details:

$$\begin{aligned}
& \sum_{n=1}^N \pi_n \left\{ (\psi_n - \lambda) \left[U_1 + \sum_{s=1}^{n-1} \left(v\left(\frac{y_s}{\theta_s}\right) - v\left(\frac{y_{s+1}}{\theta_{s+1}}\right) \right) \right] \right\} \\
& = \pi_1 (\psi_1 - \lambda) U_1 + \pi_2 (\psi_2 - \lambda) \left[U_1 + \left(v\left(\frac{y_1}{\theta_1}\right) - v\left(\frac{y_2}{\theta_2}\right) \right) \right] + \pi_3 (\psi_3 - \lambda) \left[U_1 + \left(v\left(\frac{y_1}{\theta_1}\right) - v\left(\frac{y_2}{\theta_2}\right) + v\left(\frac{y_2}{\theta_2}\right) - v\left(\frac{y_3}{\theta_3}\right) \right) \right] + \dots \\
& = \left[\sum_{s=1}^N \pi_s (\psi_s - \lambda) \right] U_1 + \left[\sum_{s=2}^N \pi_s (\psi_s - \lambda) \right] \left[\left(v\left(\frac{y_1}{\theta_1}\right) - v\left(\frac{y_2}{\theta_2}\right) \right) \right] + \left[\sum_{s=3}^N \pi_s (\psi_s - \lambda) \right] \left[\left(v\left(\frac{y_2}{\theta_2}\right) - v\left(\frac{y_3}{\theta_3}\right) \right) \right] + \dots
\end{aligned}$$

where for each n , we isolated all terms with y_n . Then, from the definition of conditional expectation:

$$\sum_{s=1}^N \pi_s (\psi_s - \lambda) = \mathbf{E}[\psi - \lambda]; \quad \sum_{s=2}^N \pi_s (\psi_s - \lambda) = (1 - P_1) \mathbf{E}[\psi - \lambda | s > 1]; \quad \sum_{s=3}^N \pi_s (\psi_s - \lambda) = (1 - P_2) \mathbf{E}[\psi - \lambda | s > 2]; \quad \dots$$

Before taking the FOCs with respect to y_n consider again the structure of the so re-arranged objective function. If we divide by $\lambda > 0$ and rearrange, we get:

$$\sum_{n=1}^N \pi_n \left\{ y_n - v\left(\frac{y_n}{\theta_n}\right) - \frac{(1-P_n) \mathbf{E}[\lambda - \psi | s > n]}{\pi_n \lambda} \left[v\left(\frac{y_n}{\theta_n}\right) - v\left(\frac{y_n}{\theta_{n+1}}\right) \right] \right\}.$$

As emphasized by Roger Myerson, in a second best problem, the planner maximized the ‘Virtual’ surplus. That is, each component of the the objective function can be divided into two key sub-components: a first best component plus an ‘externality’ emerging from the presence of the incentive compatibility constraints:

$$\sum_{n=1}^N \pi_n \left\{ \underbrace{y_n - v\left(\frac{y_n}{\theta_n}\right)}_{\text{FB component}} - \underbrace{\frac{(1-P_n) \mathbf{E}[\lambda - \psi | s > n]}{\pi_n \lambda}}_{\text{Value of Redistribution}} \underbrace{\left[v\left(\frac{y_n}{\theta_n}\right) - v\left(\frac{y_n}{\theta_{n+1}}\right) \right]}_{\text{Information Rents}} \right\}.$$

Externality

In turn, the externality component can be divided into a component that indicates how the choice of y_n affects the information rents received by agents of higher types, and a component that indicates how much the planner dislikes giving information rents to the high types, that is the value attached by the planner to redistribution. Taking derivative w.r.t. y_n we obtain

$$\pi_n \left\{ \underbrace{1 - \frac{1}{\theta_n} v'\left(\frac{y_n}{\theta_n}\right)}_{\text{FB component}} - \underbrace{\frac{(1-P_n) \mathbf{E}[\lambda - \psi | s > n]}{\pi_n \lambda}}_{\text{Value of Redistribution}} \underbrace{\left[\frac{1}{\theta_n} v'\left(\frac{y_n}{\theta_n}\right) - \frac{1}{\theta_{n+1}} v'\left(\frac{y_n}{\theta_{n+1}}\right) \right]}_{\text{Information Rents}} \right\}. \quad (9.4)$$

If we now set (9.4) to zero, use the expression for the marginal taxes in the decentralization exercise, i.e. $T'(y) = 1 - \frac{1}{\theta} v'\left(\frac{y}{\theta}\right)$, and divide both sides by $\frac{1}{\theta} v'\left(\frac{y}{\theta}\right) = 1 - T'(y)$, we get

$$\frac{T'(y_n^*)}{1 - T'(y_n^*)} = \frac{1 - P_n}{\pi_n} \frac{\mathbf{E}[\lambda - \psi | s > n]}{\lambda} \left[1 - \frac{\theta_n}{\theta_{n+1}} \frac{v'\left(\frac{y_n^*}{\theta_{n+1}}\right)}{v'\left(\frac{y_n^*}{\theta_n}\right)} \right]. \quad (9.5)$$

The expression to the left hand side is monotone increasing in the marginal tax rate $T'(y) \in [0, 1]$.⁴ The right-hand side is positive, confirming that $T'(y_n^*) > 0$ (positive marginal taxes).

Ceteris paribus, the tax rate increases in $\frac{1-P_n}{\pi_n} \frac{\mathbf{E}[\lambda - \psi | s > n]}{\lambda}$. The ratio $\frac{1-P_n}{\pi_n}$ represents the relative fraction of agents of types higher than θ_n with respect to the fraction of agents of type θ_n . That is, the relative numerosity of types that would see their information rents affected if we were to alter the production of type θ_n . The term $\frac{\mathbf{E}[\lambda - \psi | s > n]}{\lambda}$ can be rewritten as $1 - \mathbf{E}\left[\frac{\psi}{\lambda} | s > n\right] \in (0, 1]$, and it increases with the relative social value of funds λ , compared to the expected value of the Pareto-weights of types above n .

When v is a power function - $v\left(\frac{y}{\theta}\right) = \frac{1}{\gamma} \left(\frac{y}{\theta}\right)^\gamma$, $\gamma > 1$ - the last term is independent on y_n , and it takes the value: $1 - \left(\frac{\theta_n}{\theta_{n+1}}\right)^\gamma \in (0, 1)$. Ceteris paribus, for any given value of the ratio $\frac{\theta_n}{\theta_{n+1}} < 1$, the larger the value of γ the higher the marginal tax. This is intuitive since - as we will see - γ is inversely related to the elasticity of labor supply.

⁴Note indeed that as $T'(y)$ increases the numerator increases and the denominator decreases.

9.4 Euler's Feasible Perturbations

Consider again the original relaxed problem:

$$\begin{aligned}
& \max_{\{y_n, c_n\}_{n=1}^N} \sum_{n=1}^N \pi_n \left[\psi_n \left(c_n - v \left(\frac{y_n}{\theta_n} \right) \right) + \lambda (y_n - c_n) \right] - \lambda G \\
\text{s.t. } & c_n - v \left(\frac{y_n}{\theta_n} \right) \geq c_{n-1} - v \left(\frac{y_{n-1}}{\theta_n} \right) & (LDIC_n) & \quad \forall n \in \{2, \dots, N\} \\
& y_n \geq y_{n-1} & (M_n) & \quad \forall n \in \{2, \dots, N\} \\
& c_n - v \left(\frac{y_n}{\theta_n} \right) \geq 0 & (IR_n) & \quad \forall n \in \{1, 2, \dots, N\}
\end{aligned}$$

Using Euler's Perturbations approach we can find different proofs for previous results and give new interpretations to the taxation formula (9.5). The idea behind this method is not very different from many of the proofs we have already seen: we know that, at optimum, we should not be able to find a feasible perturbation (i.e. a feasible change) in our choice variables that increases the value of the objective function. Typically, such perturbations are local, in the sense that they are 'small'. This is done mainly because in most cases, this is how you guarantee feasibility of the perturbed contract.

Average value of the Pareto weights. Recall that from the FOCs with respect to U_1 , taking into account the IR_1 constraint, we have already shown that:

$$\mathbf{E}[\psi] := \sum_{n=1}^N \pi_n \psi_n \leq \lambda.$$

As you may recall, this means that the weighted sum of the Pareto weight across all agents must be lower than the cost of funds. We can now see a different argument that shows this result.

Start from any feasible allocation, and consider a perturbation where consumption is uniformly (i.e. for all types and by the same amount) increased by a small amount $\Delta > 0$. Note that this perturbation is feasible as IR are satisfied a fortiori and LDIC are left untouched by *uniform changes in consumption*. The cost in terms of budget for this variation is $\lambda\Delta$. However, since the utility of each agent also increases by Δ , there is a positive effect on the value function of $\sum_{n=1}^N \pi_n \psi_n \Delta = \Delta \mathbf{E}[\psi]$. Therefore the net effect of our perturbation is:

$$\mathbf{E}(\psi - \lambda)\Delta.$$

Clearly at optimum such value cannot be positive and thus we obtain the desired result. [Q: Why cannot be sure that a *uniform decrease* in consumption is always feasible?]

Sub-optimality of first best quantities. We can also use the perturbations to show that *first best income quantities cannot be optimal* as they allow for a feasible perturbation that increases the objective function. First of all recall that there is nothing wrong in terms of feasibility with the first best income values $\{y_n^{fb}\}$. We can always find consumption values such that they are both incentive compatible and satisfying the IR constraints. The reason why

the planner does not like the first best production quantities is that they imply a too high information rents for the high types, a generalisation to the case we saw for the wine seller monopolist.

So, let us start from the first best production quantities y_s^{fb} for all s , and choose consumption in such a way that the $LDIC$'s are binding, and IR_1 and BC are satisfied (we know that IR_s , $s > 1$ will be satisfied as well. Call such level of consumption c_s^{fb} .

Consider now the following perturbation: decrease marginally both income and consumption for a generic n in such a way that U_n remains constant. Note: this is a movement along agent n 's indifference curve. In addition, decrease all c_s with $s > n$ by the same amount such that and the $LDIC_{n+1}$ binds again after the perturbation, while keeping production of high types constant at their first best levels: $y_s = y_s^{fb}$ for all $s > n$.

Recall that the equation describing agent's n indifference curve in the (y, c) plane is:

$$c_n(y) = \bar{U} + v\left(\frac{y}{\theta_n}\right),$$

and thus the slope at $\bar{y}$ is:

$$\frac{\partial c_n(\bar{y})}{\partial y} = \frac{1}{\theta_n} v'\left(\frac{\bar{y}}{\theta_n}\right).$$

By the efficiency of the first best quantities, we have:

$$\frac{\partial c_n(y_n^{fb})}{\partial y} = 1.$$

That is, at first best quantities the slope of the indifference curve in the (y, c) plane equals one. This in turn implies that we can have the same (marginal) decrease in both c_n^{fb} and y_n^{fb} while keeping the utility of agent θ_n constant. Denote by $(\hat{y}_n, \hat{c}_n)$ the perturbed values. If small, such perturbation would also keep the BC satisfied. In other terms, because the slope of the agent's indifference curve is one in a first best allocation, there are no first order losses in production efficiency, so no budget costs.

To ensure $LDIC_{n+1}$ also binds after the perturbation, the consumption perturbation $\hat{c}_{n+1} = c_{n+1}^{fb} - \Delta$ must be such that Δ solves:

$$c_{n+1}^{fb} - \Delta - v\left(\frac{y_{n+1}^{fb}}{\theta_{n+1}}\right) = \hat{c}_n - v\left(\frac{\hat{y}_n}{\theta_{n+1}}\right).$$

Note that $\Delta > 0$ since $\theta_{n+1} > \theta_n$ [why?]. Since the consumption for all types above $n + 1$ also decreases by the same amount, all the $LDIC_n$ will be satisfied and so the perturbation is incentive feasible. As we said, this perturbation does not change type n welfare (and in the relaxed problem we ignore the upwards constraints). We have hence obtained a feasible perturbation. Let's compute the value for the planner of such perturbation assuming that the resources saved by decreasing high type consumption is redistributed equally across *all* agents.⁵

Welfare cost for high types are positive. In the eyes of the planner, these costs (due to the decrease in consumption) equal $\sum_{s=n}^N \pi_s \psi_s \Delta = \Delta \cdot \mathbf{E}[\psi | s > n](1 - P_n)$. At the same time, such change in consumption to high types generates budget saving equal to $\Delta \cdot [1 - P_n]$. As we said, the resources saved through the perturbation are redistributed *uniformly across all agents*,

⁵Note again that such uniform redistribution preserved incentive compatibility.

without touching production quantities. This keeps BC, IR and IC satisfied. The welfare gain obtained from such redistribution is $\Delta \cdot (1 - P_n)E[\psi]$. This is so since we increase the utility to all agents. Overall, we have a net positive gain because the decreasing nature of ψ_n implies:⁶

$$\mathbf{E}[\psi] - \mathbf{E}[\psi|s > n] > 0.$$

Thus we have found a feasible perturbation that increases the objective function. Hence, the first best quantities cannot be optimal in presence of incentive constraints: by distorting away from efficiency we can gain in redistribution.

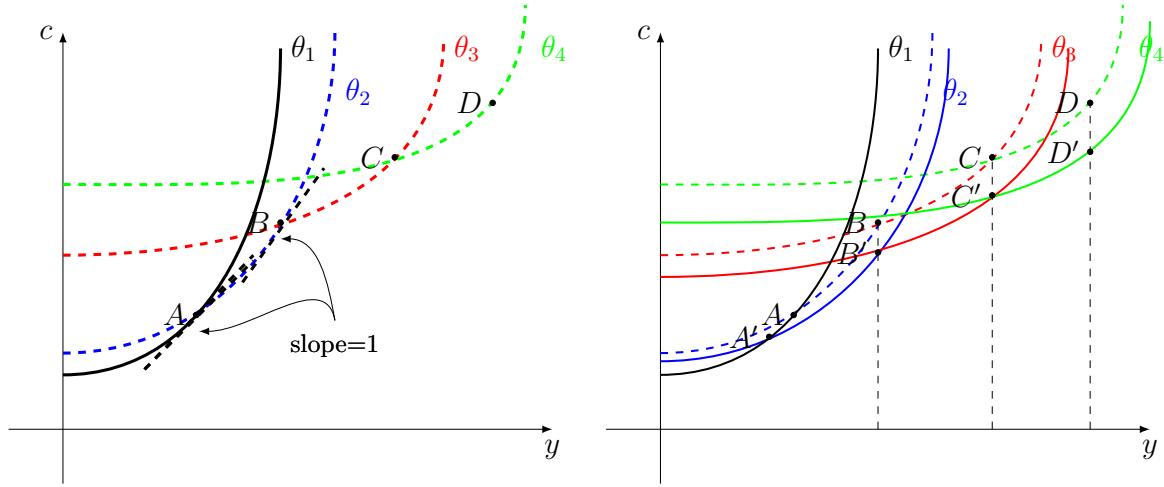


Figure 9.5: Graphical representation of the perturbation from the first best allocation with incentive compatibility, for $N = 4$. In the left panel we present the first best production quantities and consumption levels so that to satisfy the incentive compatibility constraints. In the right panel, we show an example of how the allocation would change if we were to perturb (distort downwards) type θ_1 allocation by moving along his indifference curve. This is indicated by the change from A to A' . This reduction in y_1 from the first best implies that we can decrease consumption of all higher types while still satisfy the incentive compatibility constraints, and keeping $y_n = y_n^{fb}$ for all $n > 1$. These changes are represented by the vertical movements from B, C, D to, respectively, B', C', D' .

In the left panel of Figure 9.5, we modified the first best allocation from Figure 9.2, by increasing consumption of higher types so that to satisfy the incentive compatibility constraints while keeping quantities at their first best levels. So at all points A, B, C and D the associated indifference curves have unitary slope as required by the efficiency condition for production. The figure in the right panel shows an example of how the allocation would change if we were to perturb (distort downwards) type θ_1 allocation by moving along his indifference curve. This is indicated by the change from A to A' . This reduction in y_1 from the first best implies that we can decrease consumption while still satisfy the incentive compatibility constraints of all higher types, and keeping $y_n = y_n^{fb}$ for all $n > 1$. These changes are represented by the vertical movements from B, C, D to, respectively, B', C', D' .

⁶The net effect is a transfer from high to low types.

9.4.1 Interpretation of the Tax Formula

We now use the perturbation method to re-derive the tax formula associated to the second best allocation. More precisely, we now show how a perturbation such as the one defined above delivers the second best trade offs. Let us start from the *second best allocation* and consider a marginal change in both y_n^* and c_n^* so that U_n is constant. We have:

$$\Delta c_n = \frac{1}{\theta_n} v' \left(\frac{y_n^*}{\theta_n} \right) \Delta y_n$$

By our previous discussion on downwards distortions, we have:

$$\frac{1}{\theta_n} v' \left(\frac{y_n^*}{\theta_n} \right) < 1,$$

and thus now:

$$|\Delta c_n| < |\Delta y_n|.$$

Consider again a change in consumption levels to types higher than θ_n with no change in production quantities. To keep $LDIC_{n+1}$ satisfied with equality, the perturbation must satisfy:⁷

$$\begin{aligned} \Delta c_{n+1} &= \Delta c_n - \frac{1}{\theta_{n+1}} v' \left(\frac{y_n^*}{\theta_{n+1}} \right) \Delta y_n \\ &= \Delta c_n - \frac{\frac{1}{\theta_{n+1}} v' \left(\frac{y_n^*}{\theta_{n+1}} \right)}{\frac{1}{\theta_n} v' \left(\frac{y_n^*}{\theta_n} \right)} \Delta c_n \\ &= \left[1 - \frac{\frac{1}{\theta_{n+1}} v' \left(\frac{y_n^*}{\theta_{n+1}} \right)}{\frac{1}{\theta_n} v' \left(\frac{y_n^*}{\theta_n} \right)} \right] \Delta c_n. \end{aligned}$$

By the same arguments made above, this perturbation leaves all LDIC satisfied as long as we modify by Δc_{n+1} all consumption level for types $s > n + 1$. As a last note, we ignore all IR constraints because only the one for the bottom type can be relevant and utility of agent θ_n is kept constant by construction.

Now, the change in consumption constructed above generates changes in the budget funds for a total social value of:

$$-\lambda(1 - P_n) \Delta c_{n+1}.$$

At the same time, the change in total welfare caused by change in consumption on the types above n equals:

$$\sum_{s=n+1}^N \pi_s \psi_s \Delta c_{n+1}.$$

⁷It is immediate to see that c_{n+1} moves in the same direction as c_n since:

$$1 - \frac{\frac{1}{\theta_{n+1}} v' \left(\frac{y_n^*}{\theta_{n+1}} \right)}{\frac{1}{\theta_n} v' \left(\frac{y_n^*}{\theta_n} \right)} > 0,$$

because $\theta_{n+1} > \theta_n$ and hence $v' (y_n^*/\theta_{n+1}) < v' (y_n^*/\theta_n)$ due to the convexity of v .

Hence, the overall net effect induced by the changes in consumption to types $n + 2$ and above equals:

$$- \left[\lambda - \frac{\sum_{s=n+1}^N \pi_s \psi_s}{1 - P_n} \right] (1 - P_n) \Delta c_{n+1}.$$

Realizing that $\left[\lambda - \frac{\sum_{s=n+1}^N \pi_s \psi_s}{1 - P_n} \right] = \mathbf{E}[\lambda - \psi | s > n]$, replacing Δc_{n+1} with $\left[1 - \frac{\frac{1}{\theta_{n+1}} v' \left(\frac{y_n}{\theta_{n+1}} \right)}{\frac{1}{\theta_n} v' \left(\frac{y_n}{\theta_n} \right)} \right] \Delta c_n$, we obtain the following expression for the net effect of the consumption changes to types $s > n$:

$$- \mathbf{E}[\lambda - \psi | s > n] (1 - P_n) \left[1 - \frac{\frac{1}{\theta_{n+1}} v' \left(\frac{y_n}{\theta_{n+1}} \right)}{\frac{1}{\theta_n} v' \left(\frac{y_n}{\theta_n} \right)} \right] \Delta c_n. \quad (9.6)$$

We should now account for the effects on the initial consumption variation to the allocation to agent θ_n . This is no different from zero since - differently from the case where we start from the first best - the marginal changes in consumption and income that leaves agent n indifferent do not cancel out. This is the case since the slope of the indifference curve of the agent in the second best is in general different from zero. Hence, in terms of the value of the budget resources the impact of the perturbation to agent θ_n equals:

$$\begin{aligned} \lambda \pi_n [\Delta y_n - \Delta c_n] &= \lambda \pi_n \left[\frac{1}{\frac{1}{\theta_n} v' \left(\frac{y_n}{\theta_n} \right)} - 1 \right] \Delta c_n \\ &= \lambda \pi_n \left[\frac{1 - \frac{1}{\theta_n} v' \left(\frac{y_n}{\theta_n} \right)}{\frac{1}{\theta_n} v' \left(\frac{y_n}{\theta_n} \right)} \right] \Delta c_n \\ &= \lambda \pi_n \left[\frac{T'(y_n)}{1 - T'(y_n)} \right] \Delta c_n. \end{aligned} \quad (9.7)$$

We are almost done. Note now that, by definition, *a second best contract cannot allow for feasible perturbations that improves on the objective function*. The two effects should hence be equalized. Equalizing the quantity in (9.6) with the last row in (9.7), delivers:

$$\lambda \pi_n \left[\frac{T'(y_n)}{1 - T'(y_n)} \right] \Delta c_n = \mathbf{E}[\lambda - \psi | s > n] (1 - P_n) \left[1 - \frac{\frac{1}{\theta_{n+1}} v' \left(\frac{y_n}{\theta_{n+1}} \right)}{\frac{1}{\theta_n} v' \left(\frac{y_n}{\theta_n} \right)} \right] \Delta c_n.$$

After dividing both sides by $\lambda \pi_n \Delta c_n$ we obtain our original tax formula (9.5):

$$\frac{T'(y_n)}{1 - T'(y_n)} = \frac{(1 - P_n)}{\pi_n} \frac{\mathbf{E}[\lambda - \psi | s > n]}{\lambda} \left[1 - \frac{\frac{1}{\theta_{n+1}} v' \left(\frac{y_n}{\theta_{n+1}} \right)}{\frac{1}{\theta_n} v' \left(\frac{y_n}{\theta_n} \right)} \right].$$

9.4.2 A interpretation of the Elasticity

With some algebra we can give a labor economics interpretation to the last term in the tax formula (9.5), with an empirical bite. Consider first the following:

$$\begin{aligned}
 1 - \frac{\theta_n}{\theta_{n+1}} \frac{v' \left(\frac{y_n^*}{\theta_{n+1}} \right)}{v' \left(\frac{y_n^*}{\theta_n} \right)} &= \frac{\frac{d}{dy} \left[v \left(\frac{y_n^*}{\theta_n} \right) - v \left(\frac{y_n^*}{\theta_{n+1}} \right) \right]}{\frac{1}{\theta_n} v' \left(\frac{y_n^*}{\theta_n} \right)} \\
 &= \frac{\text{change in } c_{n+1} \text{ allowed (info rents of } \theta_{n+1}) \text{ by the change in } y_n}{\text{change in } c_n \text{ (for } \theta_n) \text{ implied by a change in } y_n \text{ keeping } U_n \text{ fix}}
 \end{aligned}$$

This indicates an alternative meaning of the last term. Let us now go a bit further by approximating the formula. For θ_n close to θ_{n+1} , we have:

$$\begin{aligned}
 \frac{\frac{d}{dy} \left[v \left(\frac{y_n^*}{\theta_n} \right) - v \left(\frac{y_n^*}{\theta_{n+1}} \right) \right]}{\frac{1}{\theta_n} v' \left(\frac{y_n^*}{\theta_n} \right)} &\approx \frac{\frac{d^2}{dy d\theta} \left[v \left(\frac{y_n^*}{\theta_n} \right) \right] (\theta_n - \theta_{n+1})}{\frac{1}{\theta_n} v' \left(\frac{y_n^*}{\theta_n} \right)} = - \frac{\frac{d}{d\theta} \left[\frac{1}{\theta_n} v' \left(\frac{y_n^*}{\theta_n} \right) \right] (\theta_{n+1} - \theta_n)}{\frac{1}{\theta_n} v' \left(\frac{y_n^*}{\theta_n} \right)} \\
 &= \frac{\left[\frac{1}{\theta_n^2} v' \left(\frac{y_n^*}{\theta_n} \right) + \frac{y_n^*}{\theta_n^3} v'' \left(\frac{y_n^*}{\theta_n} \right) \right] (\theta_{n+1} - \theta_n)}{\frac{1}{\theta_n} v' \left(\frac{y_n^*}{\theta_n} \right)} \\
 &= \frac{[v'(\ell_n^*) + \ell_n^* v''(\ell_n^*)] (\theta_{n+1} - \theta_n)}{\theta_n v'(\ell_n^*)} = \left[1 + \frac{1}{\varepsilon_n^*} \right] \frac{\theta_{n+1} - \theta_n}{\theta_n},
 \end{aligned}$$

where the elasticity of labour supply $\varepsilon_n^* := \frac{v'(\ell_n^*)}{\ell_n^* v''(\ell_n^*)}$ corresponds to the change in log hours ℓ_n when the log of net wage $w_n := \theta_n(1 - T'(\cdot))$ changes. They are hence percentage changes, i.e. elasticities.

Hence, the distortion turns to be inversely proportional to the elasticity of the labor supply. When v takes the power form, labor supply elasticity is constant. For $v(\ell) = \frac{\ell^{1+\gamma}}{1+\gamma}$ with $\gamma > 0$ we indeed have:

$$\varepsilon = \frac{v'(\ell)}{\ell v''(\ell)} = \frac{\ell^\gamma}{\gamma \ell \ell^{\gamma-1}} = \frac{1}{\gamma}.$$

and hence $1 + \frac{1}{\varepsilon} = 1 + \gamma$.

Interpretation of the Tax Formula again. To conclude, we summarize our interpretations to the different components of the tax formula. Let's write it as follows:

$$\frac{T'(y_n^*)}{1 - T'(y_n^*)} = \frac{(1 - P_n)}{\pi_n} \frac{\mathbf{E}[\lambda - \psi | s > n]}{\lambda} \left[1 + \frac{1}{\varepsilon_n^*} \right] \frac{\theta_{n+1} - \theta_n}{\theta_n}.$$

We have:

- $\frac{1 - P_n}{\pi_n} \frac{\mathbf{E}[\lambda - \psi | s > n]}{\lambda}$: The marginal tax on agent θ_n is large when the value of the redistribution from agents above him is large and when the relative numerosity of agents above n is large compared to the numerosity of agent θ_n (π_n).
- $1 + \frac{1}{\varepsilon_n^*}$: The marginal tax is low when agent θ_n has very elastic labor supply. An increase in tax reduces labor supply of type θ_n proportional to ε_n^* . If ε_n^* is large, an increase in tax implies a large reduction of y_n . As a consequence, such change only allows for a relatively small decrease in c_{n+1} , hence a small redistributive gain.

- $\frac{\theta_{n+1}-\theta_n}{\theta_n}$: Typically assumed equal steps (continuum) $\theta_{n+1} - \theta_n \equiv \kappa$. If θ_n is large, a decrease in labor supply of agent θ_n induces a large effect on budget. And hence it is not very desirable.⁸

9.5 How much Tax on Top Income Earners?

Recall the tax formula assuming $\theta_{n+1} - \theta_n = 1$

$$\frac{T'(y_n^*)}{1 - T'(y_n^*)} = \frac{1 - P_n}{\theta_n \pi_n} \frac{\mathbf{E}[\lambda - \psi | s > n]}{\lambda} \left[1 + \frac{1}{\varepsilon_n^*} \right]$$

We saw that if $N < \infty$ then since $P_N = 1$ we must have $T'(y_N^*) = 0$. Consider now an unbounded support for θ , in our case, with equal spacing, this would correspond to $N \rightarrow \infty$.

What happens to $\lim_{n \rightarrow \infty} \frac{1 - P_n}{\theta_n \pi_n}$? Empirically, it is very difficult to know. Econometricians are debating between the Log Normal or the Pareto distribution. For example, Maddala suggest the following:

'The Pareto function fits the data fairly well towards the higher levels but the fit is poor towards the low income levels. The lognormal fits the lower income levels better but its fit towards the upper end is far from satisfactory.' Singh & Maddala (1976)

But the assumption on the limit distribution matters a lot for optimal taxation. For example, let's consider the **Pareto Distribution**. This is also denoted as *Type I Pareto distribution*. The cumulate of such distribution is:

$$P(\theta) = 1 - \left(\frac{\theta_m}{\theta} \right)^\alpha, \quad \alpha, \theta_m > 0,$$

with support $\theta \geq \theta_m$. Within the support, the density is

$$\pi(\theta) = \frac{\alpha \theta_m^\alpha}{\theta^{\alpha+1}},$$

so the hazard rate normalized at θ is constant:

$$\frac{1 - P(\theta)}{\theta \pi(\theta)} = \frac{1}{\alpha}.$$

So

$$\frac{T'(y_n^*)}{1 - T'(y_n^*)} = \alpha \frac{\mathbf{E}[\lambda - \psi | s > n]}{\lambda} \left[1 + \frac{1}{\varepsilon_n^*} \right].$$

The tax rate converges to a positive constant (as long as ε_n^* remains bounded away from zero). That is, the marginal tax is never zero.

Here below we report graphical representation of the estimates conducted by Saez on the income distribution (figure 9.6) and of the optimal income taxes under different assumption on agent's utility and Pareto weights (figure 9.7).

⁸The argument does not depend on the spacing on Θ . With exponential spacing for example, i.e., $\frac{\theta_{n+1}-\theta_n}{\theta_n} \equiv g$, the effect of θ seem to disappear, however, the interpretation of P_n would differ as well in the continuum.

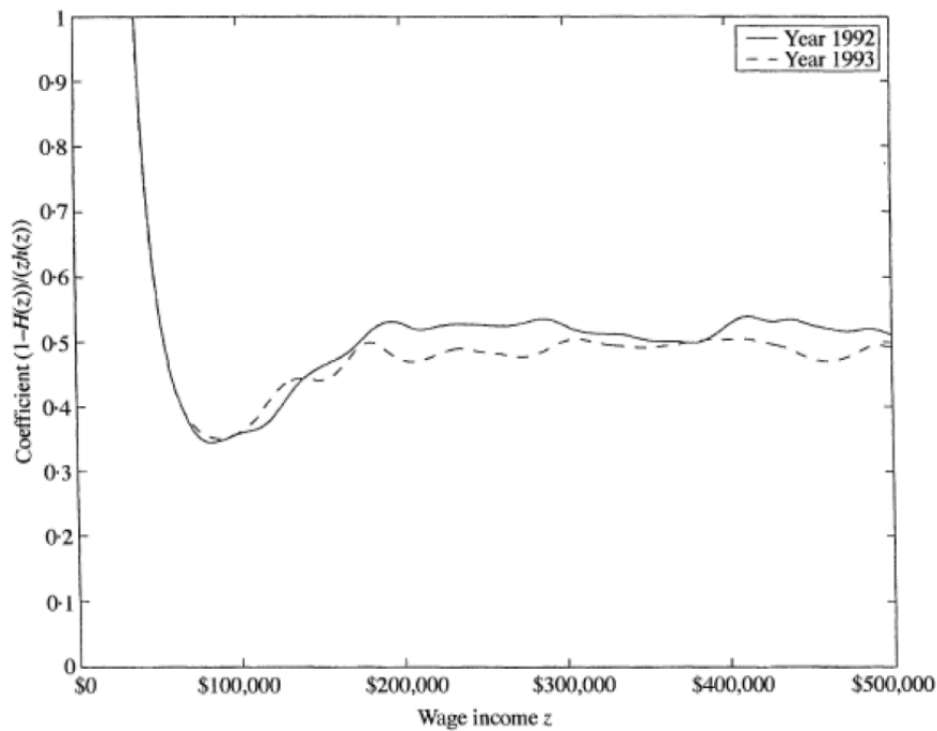


Figure 9.6: This graph reports the estimates of income distribution, in particular of the hazard rate, performed by Saez in his 2001 article published into *the Review of Economic Studies*, for the years 1992 and 1993. Note that the distribution is not reported in the space of types but on income levels, one hence need to factor in the mapping $y^*(\theta)$. Anyway, the distribution of labor income H seems to be close to Type I Pareto, or at least it seems clear that the hazard rate $\frac{1-H(y)}{yh(y)}$ tends to converge to a positive value for high income levels.

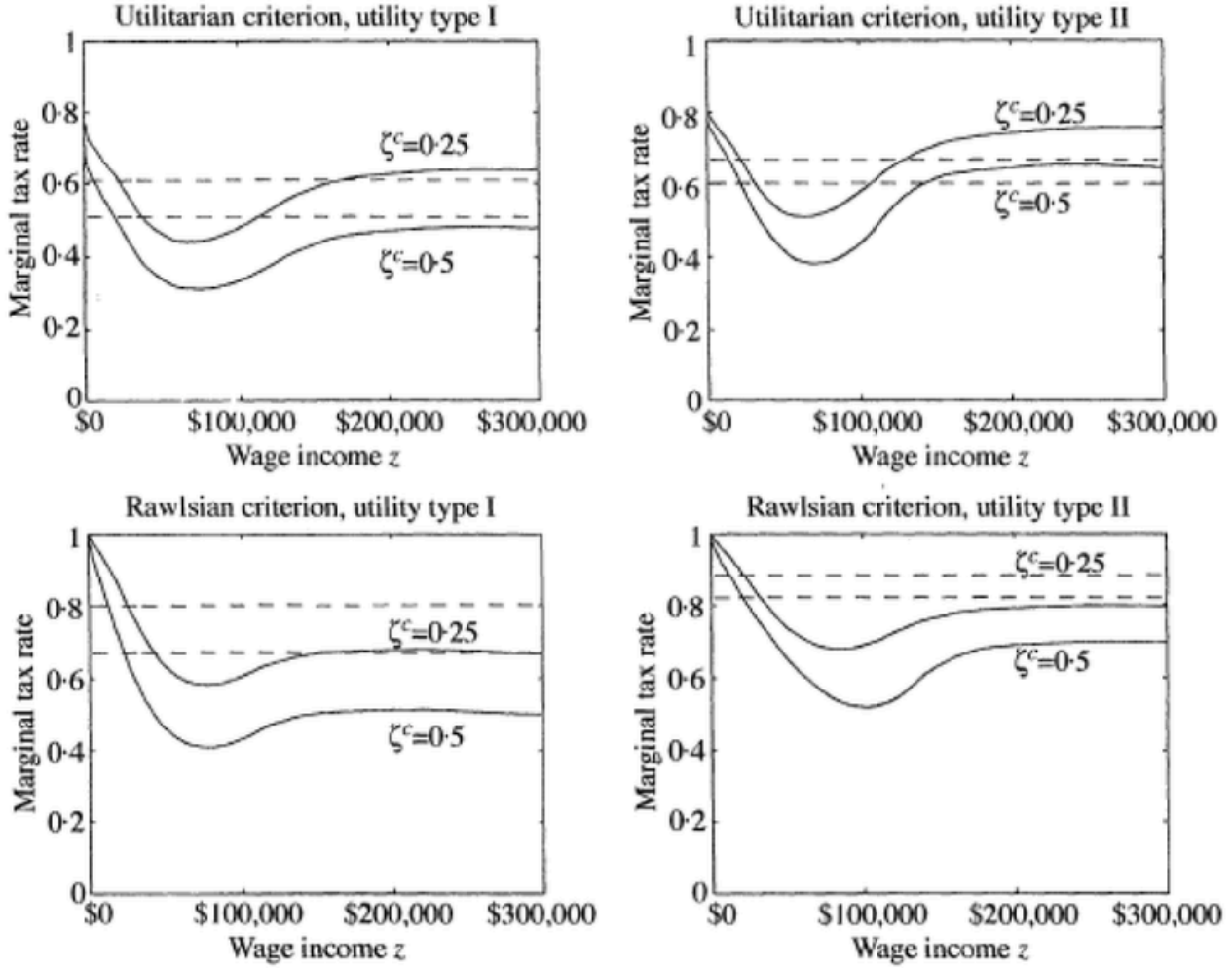


Figure 9.7: The four panels report the simulated optimal marginal tax rates as compute by Saez in his 2001 article published into *the Review of Economic Studies*. He uses the standard Mirrlees model with a continuum of types and considers different specifications for utility function and social welfare. In all graphs, ζ_c represents the compensate labor supply elasticity. The Utilitarian criterium weights all agents equally while the Rawlsian criterium corresponds to $\psi_1 > 0$ and $\psi_n = 0$ for $n > 1$ in our welfare function. Utility type I is a log transformation of our utility function: $U(c, \ell) = \ln(c - v(\ell))$, while utility type II: $U(c, \ell) = \ln c - \ln(1 + v(\ell))$. In al cases, Saez assumes $v(\ell) = \frac{\ell^\gamma}{\gamma}$, and hence $\zeta_c = \frac{1}{\gamma-1}$. As it is clear from the graphs, he obtain large optimal marginal rates for all income levels. The income distribution is as reported in figure 9.6.

9.6 Bunching (Pooling)

Recall we did not impose the monotonicity constraints M_n on income levels. If the first order condition (9.4) delivers a solution y_n^* which is increasing in n we are done. That is, although we did not impose the conditions M_n , they end up being satisfied anyway. What if they are not? Namely, what shall we do if the solution to problem (9.3) without imposing constraints M_n delivers a non-monotone schedule for income?

In this case we will have a ‘bunching solution’ for some types. Namely, each agent in the range of the bunching solution receives the same allocation. Note indeed that whenever $y_n^* = y_{n-1}^*$, the *LDIC* implies $c_n^* = c_{n-1}^*$. If we have bunching from θ_n to θ_{n-k} , denoting $\bar{y}^*$ the common income in the bunching set, solves:

$$\sum_{i=n-k}^n \pi_i \left[1 - \frac{1}{\theta_i} v' \left(\frac{\bar{y}^*}{\theta_i} \right) \right] = \sum_{i=n-k}^n [1 - P_i] \frac{\mathbf{E}[\lambda - \psi | s > i]}{\lambda} \left[\frac{1}{\theta_i} v' \left(\frac{\bar{y}^*}{\theta_i} \right) - \frac{1}{\theta_{i+1}} v' \left(\frac{\bar{y}^*}{\theta_{i+1}} \right) \right].$$

Note that this condition represents the analogous to (9.4) when more than one type is bunched together.

Note that along the bunching region, also the upwards constraints bind (trivially); in fact, one of more upwards binding constraint are typically the ‘cause’ of bunching solutions. It is hence not a surprise that we distort all agents between θ_{n-k} and θ_n and that such a distortion might be upwards (i.e. $\bar{y} > y_s^{fb}$) for some types s in the bunching set. Again, one can show that we cannot have bunching for the highest type ($n = N$) [Do it!]. Moreover, for all θ_s not belonging to any bunching set, the FOCs are exactly as before: and the solution for those types is hence exactly as discussed above. Namely, the additive structure of the problem implies that - absent income effects on labor supply and exogenously specified Pareto weights - bunching does not have consequences for the allocation to types outside the bunching set.

Ironing. An example of a bunching solution is displayed in Figure 9.8. The figure reports income (in the vertical axis) as a function of the index of θ (in the horizontal axis). Corresponding to θ_3 and θ_4 income would be lower than the level corresponding to θ_2 . Since this cannot be, we will have a bunching solution. The solution flattens the income schedule for such income levels till monotonicity is restored. This flattened segment is represented by the horizontal blue line. Such procedure is denoted in the literature as ‘ironing’ and it turns out to be very powerful in the case with a continuum of types. In this case, one can take the solution to the problem without the monotonicity constraint, draw a flat line for all ranges of non-monotonicity and so obtain the solution to the original problem.

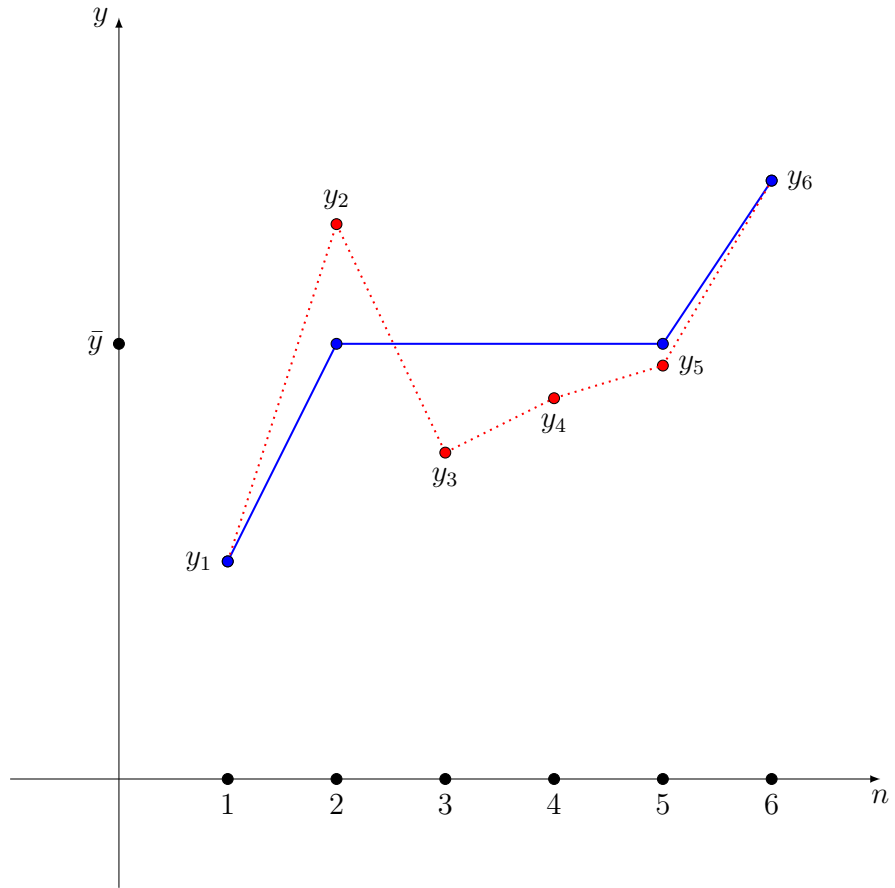


Figure 9.8: Bunching. The red dotted line represents the solution of the relaxed problem, ignoring the monotonicity condition on y_n . The solid blue line is the final solution of the second best problem, which contemplates bunching. The solutions not involved on the bunching (in this example, y_1 and y_6) do not change while the level of production $\bar{y}$ for the types in the bunching set is a sort of weighted average of the solutions to the relaxed problem (the red dots).